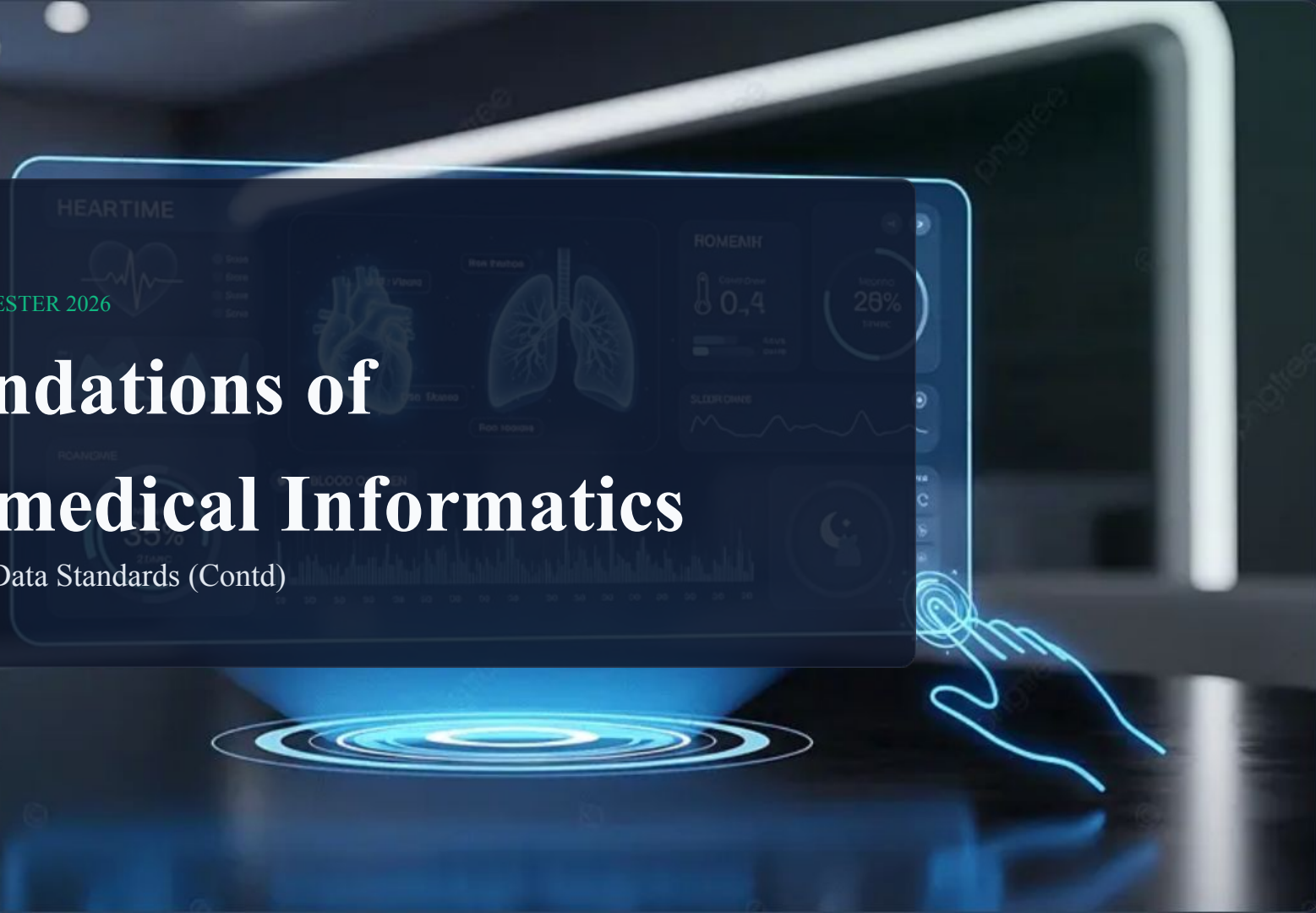


WINTER SEMESTER 2026

Foundations of Biomedical Informatics

Lecture 6: Data Standards (Contd)

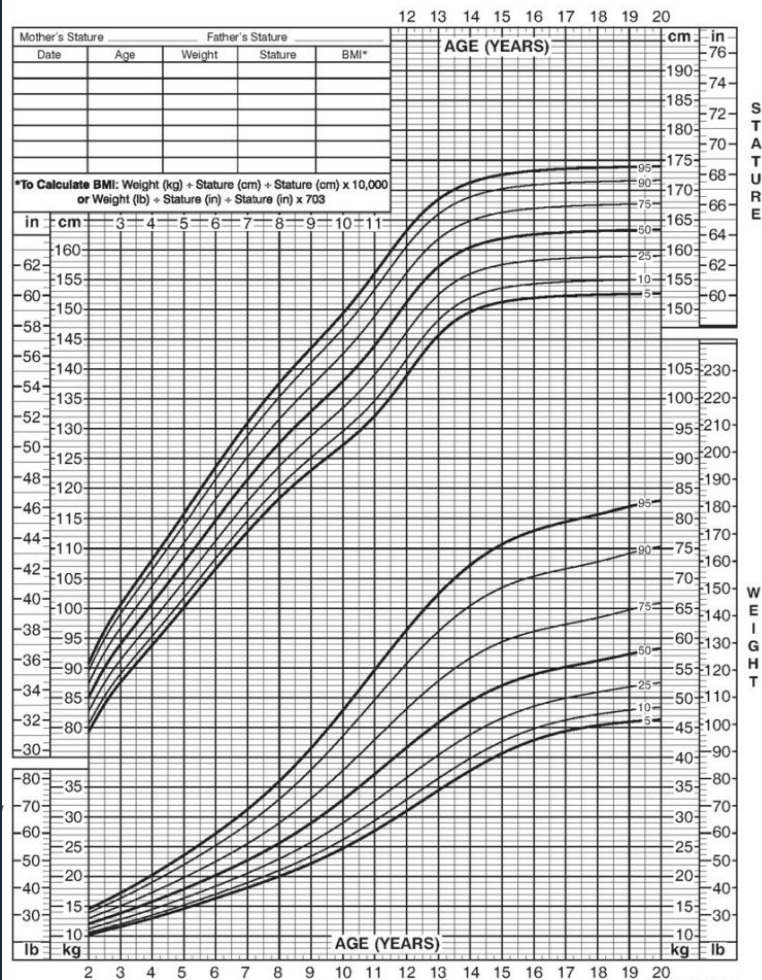


What do Standards Help With?

Healthcare involves many actors, across time, roles, and locations

- **Enables accurate communication between actors, roles, locations**
- **Standards ensure shared meaning and interpretation**
- **Comparability of data across systems and institutions**
- **Interoperability between health information systems**
- **Reusability of data for care, research, and analytics**
- **Improved efficiency of healthcare services**
- **Reduced errors by avoiding duplicated data entry**

Growth Charts



Published May 30, 2000 (modified 11/21/00).

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
<http://www.cdc.gov/growthcharts>



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Why Standards Enable Effective Communication:

The Case of Growth Charts

- **Growth charts are powerful communication tools**
 - a. They translate complex longitudinal data into intuitive visual forms
 - b. Usable by clinicians, parents, nurses, and public health professionals
- **Built on shared standards**
 - a. Standardized definitions of age, height, weight, and BMI
 - b. Uniform measurement units, reference populations, and percentiles
- **Enable common understanding**
 - a. A 50th or 95th percentile has the same meaning everywhere
 - b. Support comparability and continuity of care
- Growth can be tracked reliably across clinics, regions, and time
 - a. Enables early detection of abnormal growth patterns
- Illustrate the core value of informatics standards: Data → standardized representation → meaningful information → informed decisions

Interoperability Standards

What is Interoperability?

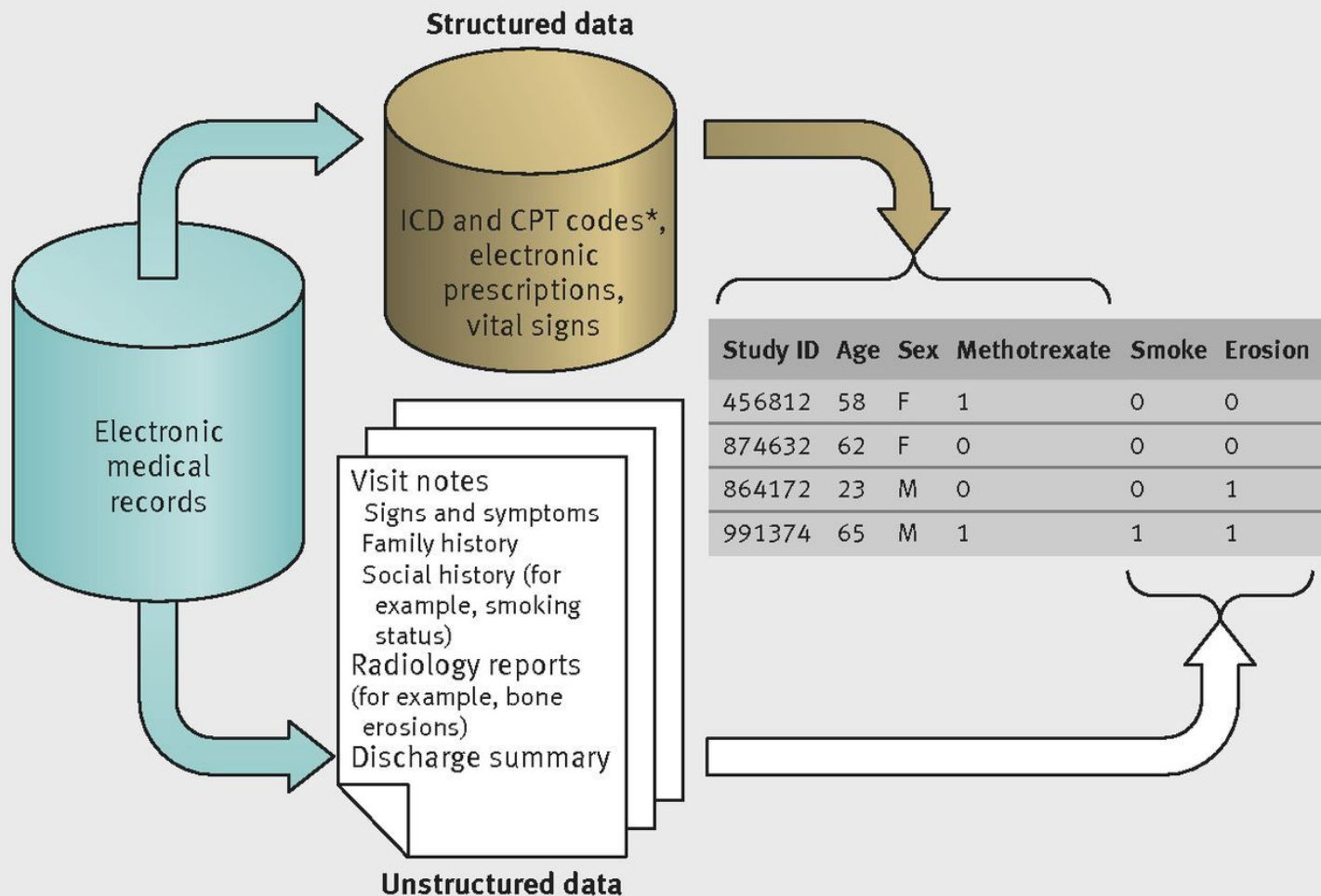
- **Interoperability is ability of two or more systems or components to exchange information and to use the information that has been exchanged (IEEE 1990)**
- **Levels of Interoperability**
 - **Technical - exchange of information**
 - **Semantic - capability of the recipient to “use” that information**
 - **Process - actual use of information**
 - **Human - clinical interoperability**

| What is an Electronic Health Record (EHR)?

ISO/DTR 20514: “A repository of information regarding the health of a subject of care in **computer processable** form, stored and transmitted **securely**, and accessible by multiple authorized users. It has a **standardized** information **model**, which is independent of EHR systems.

Its primary purpose is the support of continuing efficient and quality integrated healthcare and it contains information, which is **retrospective, concurrent, and prospective**”

EHR



Coding Systems

- The term coding system is most commonly used in the context of medical coding.
- Medical codes and coding systems are often treated as:
 - Vocabularies
 - Terminologies
- These systems are primarily designed for:
 - Clinical documentation
 - Reimbursement and billing
- A key limitation:
 - Medical coding is not optimized for analytics or data preservation
 - Terms and concepts may lack rigor or precision

Table 1. Types of Terminologies Used for Computational Analysis.*

Term	Definition
Index	List of relevant terms pulled directly from a body of unstructured or semistructured text. An index is produced to improve the speed and relevance of search results.
Terminology	Set of preferred or official terms in a domain. A terminology may be a systematic nomenclature supported by a centralized body or as simple as the common usage that arises in a specific community of practice.
Thesaurus	Terminology that clusters synonyms and plesionyms (near synonyms) into categories.
Controlled vocabulary	Set of preferred terms created specifically for a domain or body of text.
Classification	Controlled vocabulary that is intended to comprehensively describe a topic or domain from a conceptual perspective and is not developed solely from a text corpus that it is meant to describe.
Statistical classification	Classification in which all concepts are mutually exclusive to avoid counting anything twice. This is typically achieved with the use of a monohierarchy, in which each concept has one and only one parent, such as the <i>International Classification of Diseases</i> (ICD). ⁷ A statistical classification is exhaustive because it includes residual categories such as “unspecified” or “not elsewhere classified.”
Ontology	Controlled terminology invoking formal semantic relationships between and among concepts, manifested as a type of description logic, which is a subset of first-order predicate logic, chosen to accommodate computational tractability. ⁸ A common example is OWL (Web Ontology Language; www.w3.org/OWL/).

Table 2. Standards, Terminologies, and Ontologies Widely Used in Clinical Medicine.*

Type of Data	Ontology	Example of Term
Diagnoses	Systematized Nomenclature of Medicine Clinical Terms (SNOMED CT) ICD Orphanet Rare Disease Ontology (ORDO) National Cancer Institute Thesaurus (NCIT)	Triple-negative breast carcinoma (NCIT:C71732)
Phenotypic abnormalities	Human Phenotype Ontology (HPO)	Bronchopulmonary sequestration (HP:0010960)
Medications	RxNorm DrugBank ChEMBL	Panobinostat (ChEMBL483254)
Adverse reactions	Ontology of Adverse Events (OAE)	Injection-site induration (OAE:0000323)
Procedures	Medical Dictionary for Regulatory Activities (MedDRA)	Cardiac aneurysm repair (MEDDRA/10007514)
Laboratory examinations	Logical Observation Identifiers Names and Codes (LOINC)	Creatinine in serum or plasma (LOINC:2160-0)
Imaging data	Digital Imaging and Communications in Medicine (DICOM) RadLex	Periosteal cortical thinning (RID45761)

International Classification of Diseases (ICD)

ICD-10 Chapter	Code range
I Certain infectious and parasitic diseases	A00-B99
II Neoplasms	C00-D48
III Diseases of the blood and blood-forming organs and certain disorders involving the immune system	D50-D89
IV Endocrine, nutritional and metabolic diseases	E00-E90
V Mental and behavioral disorders	F00-F99
VI Diseases of the nervous system	G00-G32
VII Diseases of the eye and adnexia	H00-H59
VIII Diseases of the ear and mastoid process	H60-H95
IX Diseases of the circulatory system	I00-I99
X Diseases of the respiratory system	J00-J99
XI Diseases of the digestive system	K00-K93
XII Diseases of the skin and subcutaneous tissue	L00-L99
XIII Diseases of the musculoskeletal system and connective tissue	M00-M99

ICD for Respiratory System

Section	Code range
Acute upper respiratory infections	J00-J06
Influenza and pneumonia	J10-J18
Other acute lower respiratory infections	J20-J22
Other diseases of the upper respiratory tract	J30-J39
Chronic lower respiratory diseases	J40-J47
Lung diseases due to external agents	J60-J70
Other respiratory diseases principally affecting the interstitium	J80-J84
Suppurative and necrotic conditions of the lower respiratory tract	J85-J86
Other diseases of the pleura	J90-J94
Other diseases of the respiratory system	J95-J99

| ICD for Upper Respiratory Infections within the System

Rubric	Code
Acute nasopharyngitis [common cold]	J00
Acute sinusitis	J01
Acute pharyngitis	J02
Acute tonsillitis	J03
Acute laryngitis and tracheitis	J04
Acute obstructive laryngitis [croup] and epiglottitis	J05
Acute upper respiratory infections of multiple and unspecified sites	J06

International Classification of Diseases (ICD)

ICD is not a single monolithic standard

- **Multiple versions exist:**
 - **ICD-9**
 - **ICD-10**
- **Global usage:**
 - **Most of the world used ICD-10 for decades**
 - **United States: Decision to adopt ICD-10: 2009**
 - **Full implementation: 2015**
 - **Practical consequence:**
 - **Real-world datasets often contain both ICD-9 and ICD-10**
 - **Depends on time and location of data collection**

Interpretations of Coding May be Noisy

Limitation: Even with standards like ICD:

- **Code choice depends on local context**
- **Example: Is the code used for:**
 - Reimbursement?
 - Cause of death?
 - Clinical documentation?
- **Who entered the code:**
 - Clinician?
 - Nurse?
 - Billing specialist?
 - Pharmacist?
- **Key takeaway: Interoperability problems are rarely solved by standards alone, they are solved by understanding context, intent, and meaning.**

A case from USA (Business Model Failure)

United States: ICD for Billing vs Reality: Sepsis coding in US hospitals

- **In the US, ICD-10 codes drive reimbursement**
- **Hospitals discovered coding sepsis (A41.x) instead of infection + organ failure significantly increased reimbursement!**
- **Result:**
 - **Hospitals began coding sepsis aggressively**
 - **Epidemiological data showed a sudden rise in sepsis incidence**
- **What actually happened: Coding behavior changed, not disease prevalence**
- **Impact:**
 - **CDC and researchers struggled to interpret trends**
 - **AI models trained on claims data over-predicted sepsis risk**
- **Key Takeaway: in the US (MIMIC Data) ICD codes reflect financial and administrative intent, not pure clinical semantics.**

| Case from India

Public vs Private Hospitals: Diabetes coding in Indian hospital systems

- In many Indian EHRs ICD-10 E11.x (Type 2 Diabetes Mellitus) is used
- In public hospitals
 - Often entered by overworked clinicians
 - Used mainly for aggregate reporting
- In private hospitals:
 - Often entered by billing staff
 - Used to justify:
 - Lab panels
 - Long-term medication
- Same ICD code, but:
- Different clinical certainty
- Different intent

Case from India

- When pooled for analytics:
- Population prevalence estimates became inconsistent
- State-level NCD dashboards showed contradictory trends
- Key takeaway: Context of code entry matters as much as the code itself.

WHO ICD used for mortality statistics During COVID

- Many death certificates list “Cardiac arrest” or “Respiratory failure”
- These are modes of death, not causes
- ICD rules require Underlying cause (e.g., MI, TB, COVID)
- States differed in how ICD codes were assigned
 - National mortality analytics became inconsistent
 - WHO and ICMR explicitly flagged:
- Coding quality, not data volume, as the bottleneck

Why digital medicine depends on interoperability

Moritz Lehne , Julian Sass, Andrea Essenwanger, Josef Schepers & Sylvia Thun

npj Digital Medicine **2**, Article number: 79 (2019) | [Cite this article](#)

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Abstract

Digital data are anticipated to transform medicine. However, most of today's medical data lack interoperability: hidden in isolated databases, incompatible systems and proprietary software, the data are difficult to exchange, analyze, and interpret. This slows down medical progress, as technologies that rely on these data – artificial intelligence, big data or mobile applications – cannot be used to their full potential. In this article, we argue that interoperability is a prerequisite for the digital innovations envisioned for future medicine. We focus on four areas where interoperable data and IT systems are particularly important: (1) artificial intelligence and big data; (2) medical communication; (3) research; and (4) international cooperation. We discuss how interoperability can facilitate digital transformation in these areas to improve the health and well-being of patients worldwide.